Program Structures and Algorithms
Spring 2025

Team Details:

| NAME | NUID |
|---|---|
| **SHREYA WANISHA** | **002331937** |
| **SHRIYA PRATAPWAR** | **002309394** |
| **DHARANA KASHYAP** | **002055360** |

**GITHUB LINK**: https://github.com/PSA-Final-Project/PSA-Final-Project

**SHEET LINK**:

1) Tic-Tac-Toe Observation Charts
2) Checkers Observation Charts

## Project Video Demonstrations

1. **Full Project Demo:**
   https://drive.google.com/file/d/1N2FmsGB22_yxEspIfwPiM9DUhbaYKSvT/view?usp=sharing

2. **TicTacToe Gameplay Demo:**
   https://drive.google.com/file/d/1WMXLxa9RBpvCWiPqA28i4Ky-aIyIm2lP/view?usp=sharing

3. **Checkers Gameplay Demo:**
   https://drive.google.com/file/d/10dpuRsVw5TFtEp0RZFBaC0HuPaaieLvK/view?usp=drive_link

We Implemented two games using the Monte Carlo Tree Search Algorithm.

1) TicTacToe
2) Checkers

# TicTacToe with MCTS

Our team has completed the **backend and User Interface** implementation of Monte Carlo Tree Search (MCTS) for **TicTacToe**, along with **comprehensive unit testing** to ensure correctness and reliability across all major classes. We also **developed a benchmarking** tool that runs games with increasing MCTS iterations **(doubling from 100 to 1600)** and **records both outcome and timing data**.

## Implemented Features:

1. **3x3 Board**
   The game is played on a standard 3x3 grid.
   Each cell can be empty, or occupied by Player X or Player O.

2. **Two Players**
- Player 0: X
- Player 1: O
   Turns alternate between the two players.

3. **Initial Setup**
   The board starts empty.
   Player X always goes first by default (configurable, if needed).

4. **Move Rules**
   Each move consists of placing a mark (X or O) in an empty cell.
   Moves must be within the bounds of the board and in unoccupied cells.

5. **Valid Move Generation**
   For a given state, all unoccupied cells are returned as legal moves.
   Valid moves are represented by (row, column) coordinates.

6. **State Transition**
   The TicTacToeState.next() method applies a move and returns a new state immutably.
   The board and current player are updated accordingly.

7. **Terminal State Detection**
   A game state is terminal if:
- One player has aligned three marks in a row (horizontally, vertically, or diagonally).
- All cells are filled with no winner (draw).

8. **Winner Detection**
- If a player aligns 3 of their marks, they are declared the winner.
- If the board is full and no player has won, the result is a draw.

9. **Randomized Move Iterator (Optional)**
   Moves can be returned in random order, deterministically seeded for testing, for compatibility with MCTS or AI simulations.

10. **Difficulty Levels**
    The game supports multiple difficulty levels for AI opponents:

- **Easy**: Random valid moves.

- **Medium**: Prioritizes winning moves, blocks opponent's immediate wins.

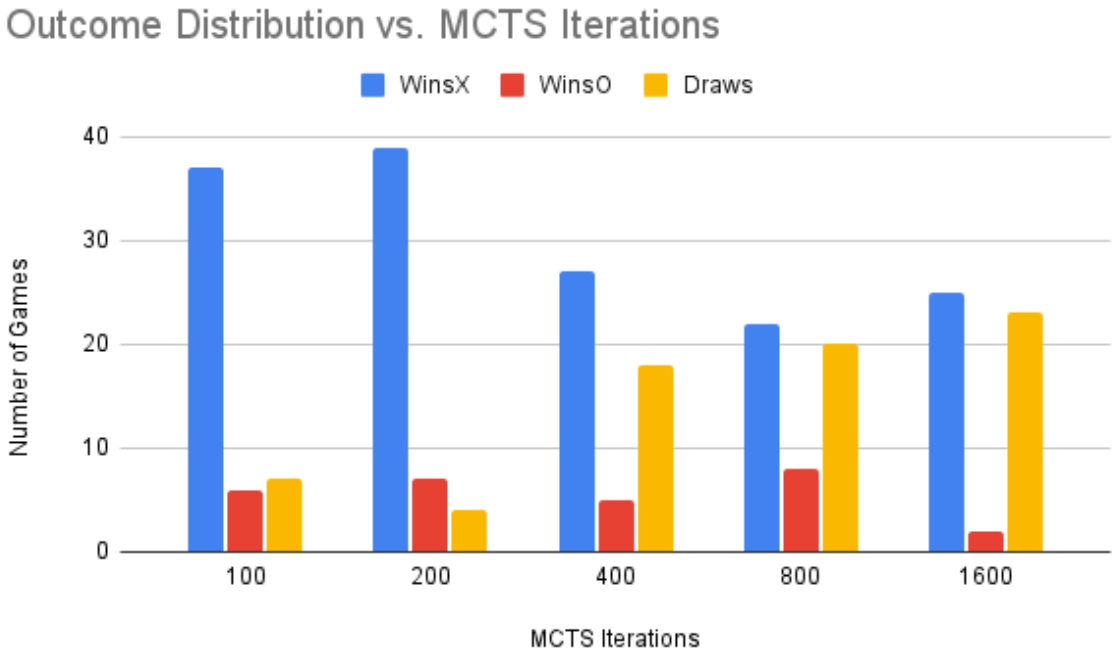
- **Hard**: Uses Minimax or MCTS to evaluate best possible moves.

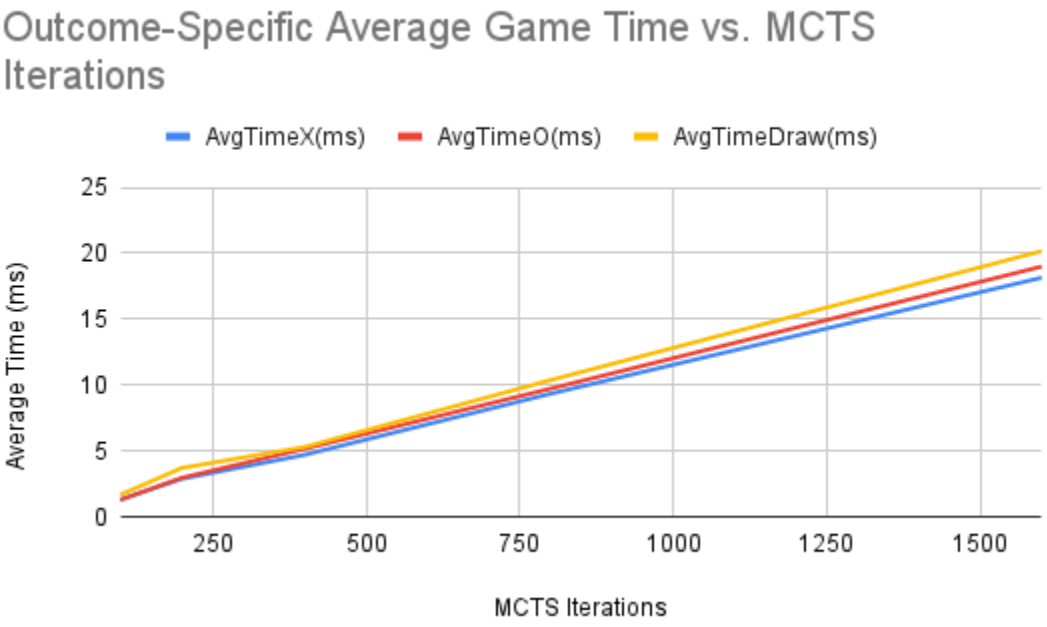
11. **MCTS Compatibility**
    Fully integrated with the interfaces: Game, State, and Move, ensuring seamless use with MCTS agents or benchmarking tools.

**Observations:**

1. **Player X dominates at 100, 200, and 800 iterations** with 50 wins.
2. **All games end in draws at 400 and 1600 iterations.**
3. **Player 0 has no wins** across all iteration levels.
4. **Average time increases** with more iterations, peaking at 1600 (13 ms).

| Iterations | AvgTotalTime(ms) | WinX | AvgTimeX(ms) | Win0 | AvgTime0(ms) | Draws | AvgTimeDraw(ms) |
|---|---|---|---|---|---|---|---|
| 100 | 3 | 50 | 3.72 | 0 | 0 | 0 | 0 |
| 200 | 5 | 50 | 5.22 | 0 | 0 | 0 | 0 |
| 400 | 4 | 0 | 0 | 0 | 0 | 50 | 4.22 |
| 800 | 7 | 50 | 7.02 | 0 | 0 | 0 | 0 |
| 1600 | 13 | 0 | 0 | 0 | 0 | 50 | 13.34 |



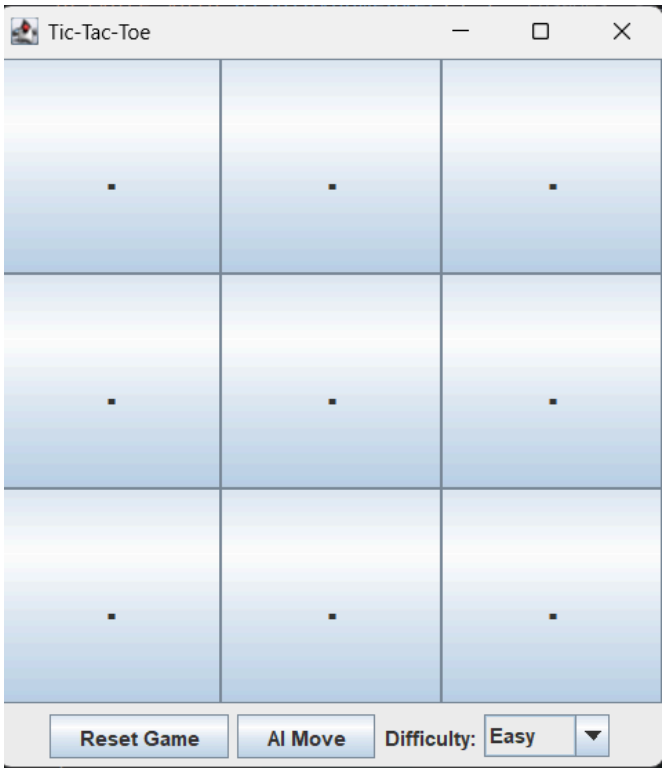

**Conclusion:**

Player X performs strongly at lower and mid iteration counts, while draws dominate at higher iterations, indicating improved defensive play. Player 0 consistently underperforms, and increased iterations lead to longer computation times.

**GUI:**

**To run the GUI, run the TicTacToeGUI.java class in the repository.**

Tic-Tac-Toe game user interface (UI) built with Java Swing. The interface consists of a 3x3 grid, where players place "X"(Human Player) and "O"(AI player) symbols to play the game.




Control Buttons (Bottom Panel):

- "Reset Game" Button: Restarts the game, clearing the board.
- "AI Move" Button: Triggers the AI to make a MCTS move.
- Dropdown for selecting difficulty level of game: Easy, Medium and Hard

# Checkers with MCTS

Our team has completed the **backend and User Interface** implementation of Monte Carlo Tree Search (MCTS) for **Checkers**, along with **comprehensive unit testing** to ensure correctness and reliability across all major classes. We also **developed a benchmarking** tool that runs games with increasing MCTS iterations **(doubling from 100 to 1600)** and **records both outcome and timing data**.

## Implemented Features:

**1. 8x8 Board**
The game is played on a standard 8x8 grid.
Only black squares (i.e., playable tiles) are used for moving pieces.

**2. Two Players**
Player 0: White
Player 1: Black
The game alternates turns between them.

**3. Initial Setup**
Each player starts with 12 pieces, placed on the first 3 rows (for White) and last 3 rows (for Black) on the black squares.

**4. Basic Movement Rules**
Each piece moves diagonally forward by one step.
White moves up the board (row--).
Black moves down the board (row++).

**5. Valid Move Generation**
For a given state and player, all legal single-step moves are generated and returned.
Only moves within board bounds and into empty target squares in the piece's forward direction are allowed.

**6. State Transition**
The CheckersState.next() method creates a new game state with the move applied (immutably).
The board and piece positions are updated accordingly.

**7. Terminal State Detection**
A game state is terminal if:
A player has no pieces left.
A player has no valid moves.

**8. Winner Detection**
If one player has no pieces or valid moves, the other player is the winner.
If both run out simultaneously or are blocked, it's a draw.

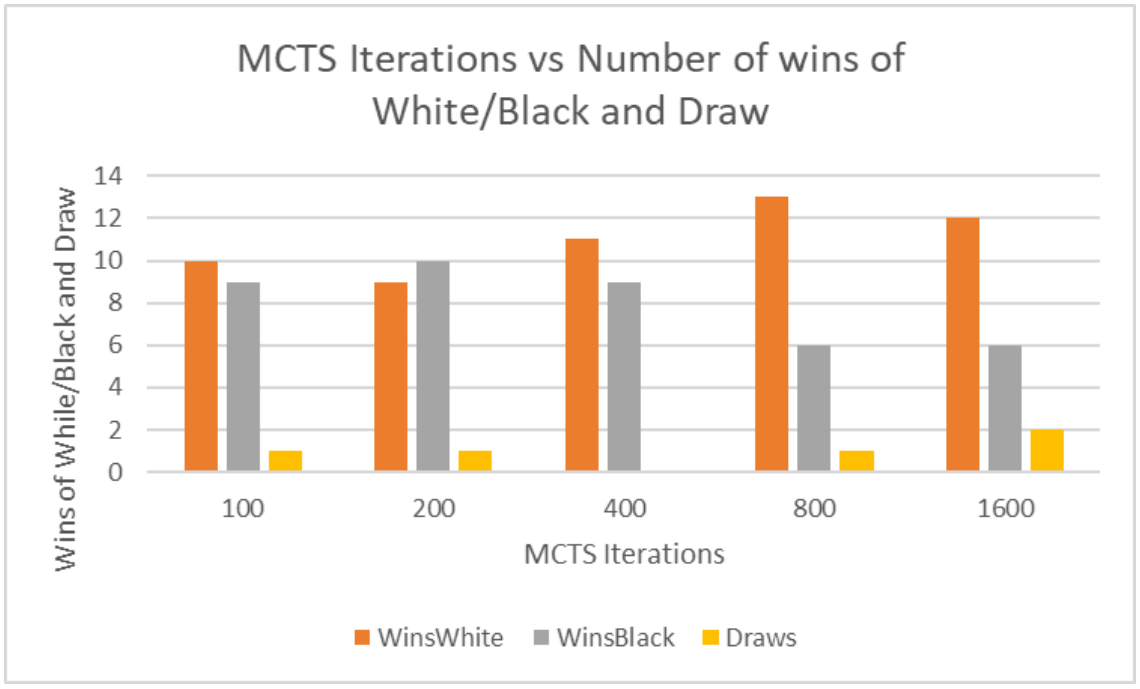
**9. Randomized Move Iterator**
Moves are randomly ordered (but deterministically for testing, using seeded Random) to comply with MCTS expectations.

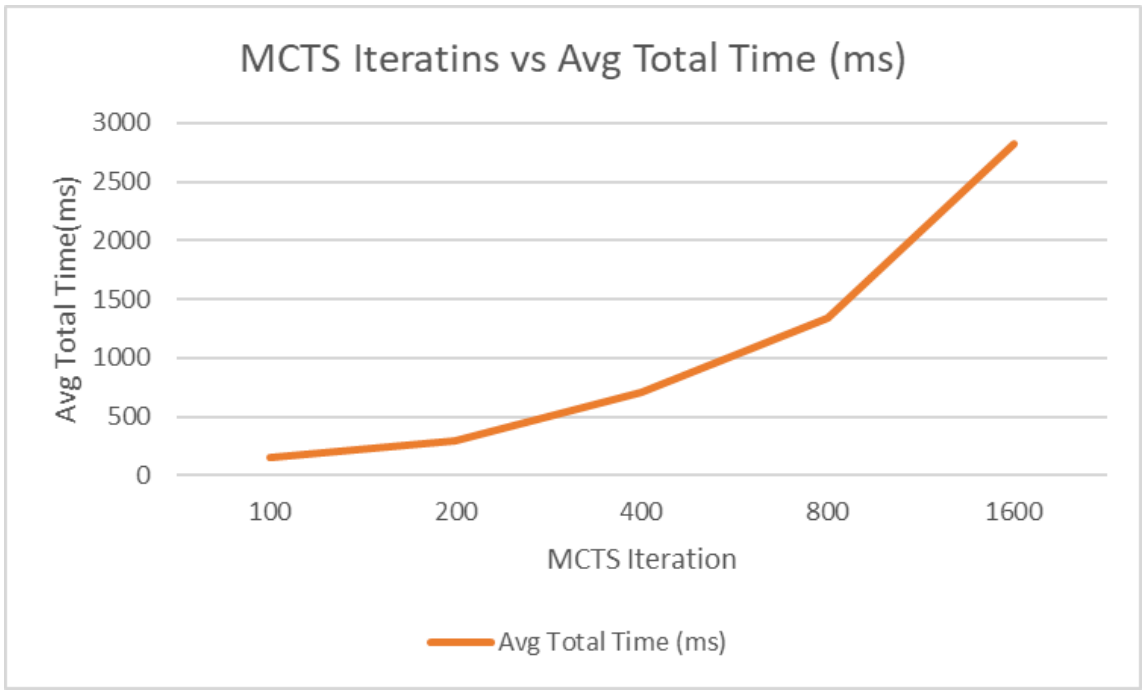
**10. MCTS Compatibility**
Fully integrated with the interfaces: Game, State, and Move.

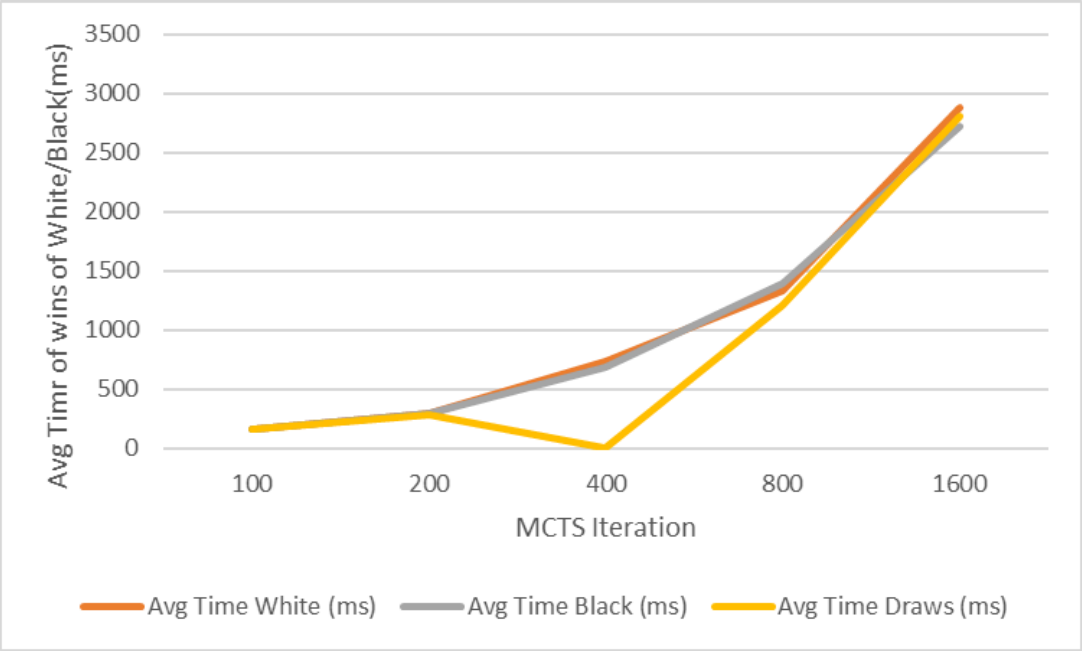
**Observations:**

1. **Higher iterations → Higher time:** Total and per-player time increase significantly with more iterations.
2. **White wins slightly more overall**, especially at 800 and 1600 iterations.
3. **Draws are rare**, occurring only at higher iterations.
4. **Performance is fairly balanced** between players, with minor variations in average time.

| Iterations | Avg Total Time (ms) | Wins White | Avg Time White (ms) | Wins Black | Avg Time Black (ms) | Draws | Avg Time Draws (ms) |
|---|---|---|---|---|---|---|---|
| 100 | 160 | 10 | 158.3 | 9 | 162.22 | 1 | 160 |
| 200 | 293 | 9 | 298.22 | 10 | 290.2 | 1 | 288 |
| 400 | 713 | 11 | 737.09 | 9 | 684.22 | 0 | 0 |
| 800 | 1342 | 13 | 1326.38 | 6 | 1398.33 | 1 | 1213 |
| 1600 | 2828 | 12 | 2886.17 | 6 | 2718.83 | 2 | 2809.5 |


MCTS Iterations vs Number of wins of White/Black and Draw
Wins of While/Black and Draw
14
12
10
8
6
4
2
0
100
200
400
800
1600
MCTS Iterations
WinsWhite
WinsBlack
Draws


MCTS Iteratins vs Avg Total Time (ms)
Avg Total Time(ms)
3000
2500
2000
1500
1000
500
0
100
200
400
800
1600
MCTS Iteration
Avg Total Time (ms)



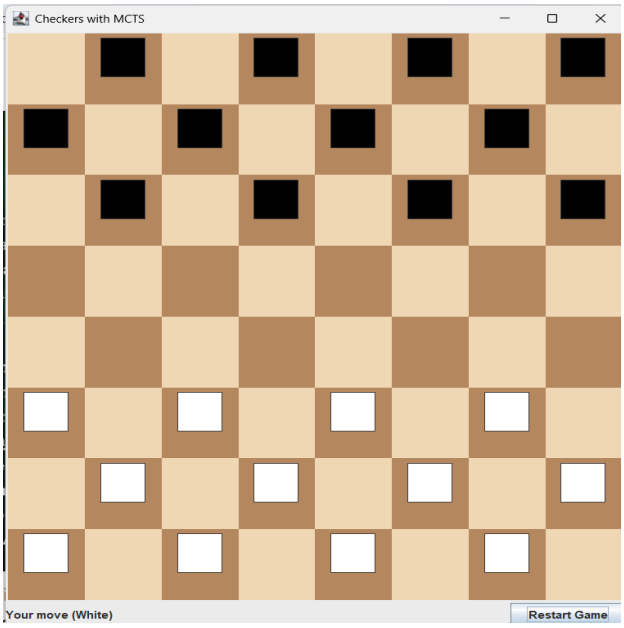

**Conclusion:**

MCTS performance improves with more iterations but at the cost of higher computation time. Around **800 iterations** offers a good trade-off between win rate and efficiency. Both players perform competitively, with **White slightly outperforming Black** overall.

## GUI:

**To run the GUI, run the CheckersGUI.java class in the repository.**

Checkers game user interface (UI) built with Java Swing. The interface consists of a 8x8 grid, where players place "White"(Human Player) and "Black"(AI player) symbols to play the game.



Control Buttons (Bottom Panel):

- "Reset Game" Button: Restarts the game, clearing the board.

# How to run the application :

Please follow the steps below to run both games locally:

1. Clone the repository:

   Open any terminal run -

   `*git clone https://github.com/PSA-Final-Project/PSA-Final-Project*`

2. Open the project using any Java IDE (e.g., IntelliJ, Eclipse, NetBeans).

**▶️ Running TicTacToe**

1. To launch the TicTacToe game UI:
   **Run** TicTacToeGUI.java
   (src/main/java/com/phasmidsoftware/dsaipg/projects/mcts/tictactoe/TicTacToeGUI.java)

2. To run the TicTacToe benchmarking tool:
   **Run** TicTacToeBenchmark.java
   (src/main/java/com/phasmidsoftware/dsaipg/projects/mcts/tictactoe/TicTacToeBenchmark.java)

**▶️ Running Checkers**

1. To launch the Checkers game UI:
   **Run** CheckersGUI.java
   (src/main/java/com/phasmidsoftware/dsaipg/projects/mcts/checkers/CheckersGUI.java)

2. To run the Checkers benchmarking tool:
   **Run** CheckersBenchmark.java
   (src/main/java/com/phasmidsoftware/dsaipg/projects/mcts/checkers/CheckersBenchmark.java)

NOTE : *Make sure Java 23+ is installed and set as your project SDK.*